PANJAB UNIVERSITY, CHANDIGARH

Outlines of Tests, syllabi and courses of reading for M. Phil. in Economics for the session 2011-12

SEMESTER – I

09 CREDITS

Paper I	Research Methodology and Quantitative Methods	03 Credits (75 marks) (60+15 internal assessment)
Paper II	Theory of Agricultural Economics	03 Credits (75 marks) (60+15 internal assessment)
	OR	
	Theory of Industrial Economics	03 Credits (75 marks) (60+15 internal assessment)
Paper III	*Econometrics –I	03 Credits (75 marks) (60+15 internal assessment)
	OR	
	Econometrics – II	03 Credits (75 marks) (60+15 internal assessment)

*Students who have not studied Econometrics at Masters level shall opt for Econometrics-I, others shall be required to opt for Econometrics-II

SEMESTER II

15 Credits

Paper IV	Economic Theory - I	03 Credits (75 marks) (60+15 internal assessment)
Paper V	Economic Theory –II	03 Credits (75 marks) (60+15 internal assessment)
Paper VI	Dissertation	09 credits (225 marks) (Dissertation 180+45 Viva-voce marks)

The students must submit the broad area(s) in which they want to do their dissertation within one month of the start of the 1st semester classes so that the process of allocation of M Phil dissertation shall commence in the 1st semester itself. There will be an internal assessment of 15 marks in each paper and would be based on Term Paper to be presented by each student.

The syllabi of the above papers is given below:

SEMESTER – I

Paper I : Research Methodology and Quantitative Methods

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hour : 40
Time : 3 Hours

OBJECTIVES: The paper aims at equipping the students with various methods and techniques which can help them to carry out research in economics.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one questions from each Unit. Each such question would carry 10 marks.

UNIT – I

Research method and methodology – Relevance and Limitations. Typology of Economic Relations, Models and Data, Procedure for Estimating Macro Level Relations (Consumption Function/Investment Function. Wage or Price Relationship).

Concepts of Fact and Theory, Essentials and Tests of a Good Theory. Nature of Causality, Logical Consistency, Verification and Falsification, Hypothesis Testing – an overview.

UNIT – II

Optimisation Techniques: Lagrangian method, Kuhn-Tucker method of inequality constraint, Linear Programming, Duality. Calculus of variation.

Matrix Algebra: Linear models, characteristic roots and vectors, linear and quadratic forms, vector differentiations.

UNIT – III

Difference Equations: Linear first order difference equations, non-linear first order equations, use of phase diagrams, second order difference equations, conditions of stability.

Game Theory: Importance of strategic behaviour in economic decision making, cooperative and non-cooperative games; different equilibrium concepts: dominant strategy equilibrium, Nash equilibrium in pure and mixed strategy; repeated games, sequential games, sub-game perfect Nash equilibrium and non-credible threats; applications: Cournot and Bertrand models, entry deterrence and commitment models.

UNIT – IV

Input-Output Model: Static and closed models; viability conditions, aggregation problem, substitution theorem, cost and price determination. Dynamic Leontief system. Balanced growth of Leontief system.

References:

Chiang, A C (1984)	:	Fundamental Methods of Mathematical Economics, (3rd Edition), McGraw Hill, Singapore.
Chiang, A C (1992)	:	Elements of Dynamic Optimization, McGraw Hill, Singapore.
Goode, W J & Hatt, P K (1952)	:	Methods of Social Research, McGraw Hill Singapore.
Koutsoyiannis, A (1978)	:	Theory of Econometrics, Macmillan Press Ltd., London.
Pindyck, Robert S. & Daniel L Rubinfeld (2009)	:	Microeconomics, Pearson Education (7th edition), India.
Varian, H R (1992)	:	Microeconomic Analysis, W W Norton and Co., (3rd edition), New York.

Paper II : Theory of Agricultural Economics

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hour : 40
Time : 3 Hours

OBJECTIVES: The paper is designed to expose the students to the contemporary issues in agricultural development both in the theory and in the Indian context.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one question from each Unit. Each such question would carry 10 marks.

UNIT – I

Agriculture Transformation – Role of Agricultural Innovations: Generations of Innovations (induced innovations, Political Economy of Public Funded Innovations, and Policies and Institutions Managing Innovations]; Technology Adoptions [Adoption and Diffusion, Risk Management, Institutional Constrains to Innovations and Farm Policy]; Ruttan-Hayami Induced Innovation Hypothesis.

Agricultural Commodity Futures and Options: Nature and Principles of Future Markets; Participants in Future Markets; Trading in the Presence of Risk Aversions and Transaction Costs, Future Prices as Forecasts.

UNIT - II

Agro-processing: Specificities of Agro-processing, Developmental Role of Agro-processing Industries, Selection and Evolution Options; Food Processing Policy of India, 2005.

Agricultural Finance: Agency Relationship, Adverse Selection, Moral Hazards and Enforcement Problem; Segmented Rural Credit Markets; Resolving Information and Incentive Problems; Managing Borrowers' Credit Risks, Agribusiness and Trade Financing and Role of Social Capital.

UNIT - III

Political Economy of Agricultural Policies: Protection across countries and sectors, influence of political institutions; collective actions of lobbing groups and strategic interactions between lobbing groups and politicians; use of inefficient instruments and rent seeking; and explaining public investment in agricultural research.

Food Security and Food Assistance: Concept, Threat, Indicators and Mechanism to Food Security; Food Assistance Programmes [both Domestic and International], and Their Effectiveness.

UNIT – IV

Indian Agriculture- Opportunities and Challenges: Expectations, Performance, Problems and Prospects of Indian Agriculture in the Post-reform Period.

Investment in Farm People: Role of Human Capital Formation [Education and Health] in Modernisation and Transition of Agrarian Economies.

References:

Bruce L. Gardner and Gordon C. Rausser (2002)	:	Handbook of Agricultural Economics, Vol-1A,1B, 2A, 2B and 3; North-Holland, Amsterdam
James G. Brown and Deloitte and Touche (1994)	:	Agro-industrial Investment and Operations, Washington, The World Bank.
Government of India (2005)	:	Food Processing Policy, Ministry of Food Processing, New Delhi.
T.W.Schultz (1981)	:	Investing in Peoples: The Economics of Population Quality, University of California, Berkley.
Radhakrishna R, S. K. Rao, S. Mahendra Dev and K Subbarao (2006)	:	India in a Globalising World: Some Aspects of Macroeconomy, Agriculture and Poverty, Academic foundation, New Delhi.
Government of India (2004)	:	State of the Indian Farmer : A Millennium Study, Academic Foundation and Ministry of Agriculture, New Delhi, (Vol. 1-27)

OR

Paper –II : Theory of Industrial Economics

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hours : 40
Time : 3 Hours

Objectives: In the 21st century, efficiency and competitiveness in all spheres of economic life and more 500 industrial sectors are at the core of success. The present syllabus is designed for the students to capture the contemporary issues in industrial development both in theory and in the Indian context.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one question from each Unit. Each such question would carry 10 marks.

UNIT – I

I. Economics of Takeovers and Mergers: Types, Objectives and Effects. Public Policy for business combinations with special reference to India: Takeover code of SEBI and Competition Policy

UNIT – II

II. Technological Progress: Concepts, Optimal Technological Change; Patents and Innovations.

UNIT - III

III. Porter's Five Competitive Forces: bargaining power of customers, the bargaining power of suppliers, the threat of new entrants, and the threat of substitute products -- combined with other variables to influence a fifth force, the level of competition in an industry. Determinants and Strategy formation.

UNIT - IV

IV. Globalization, trade liberalization and industrial market structure and economic performance: Theory and evidence since 1991.

References:

Martin, Stephen (2001)	:	Advanced Industrial Economics, Blackwell, (2nd edition).
Scherer, F M & Ross David (1990)	:	Industrial Market Structure and Economic Performance, Boston Houghton Mifflin Company, (3rd edition).
Hay, D A & Morris, D J (1979)	:	Industrial Economics and Organization Oxford Univ. Press, New York.
Shepherd, William, G, J.M. Shepherd (2003)	:	The Economics of Industrial Organization Analysis, Market Policies. Prentice Hall International, (5th edition).
Petersen H.C., W.C. Lewis (2002)	:	Managerial Economics, Prentice Hall of India, New Delhi.
Schmalensee, R and Robert Willig (1989)	:	Handbook of Industrial Organization (Vol. I & II), Elsevier, North Holland.

Paper III - Econometrics-I

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hours: 40
Time : 3 Hours

OBJECTIVES: Application of economic theory need a reasonable understanding of economic relationships and relevant statistical methods. The econometric theory thus becomes a very powerful tool for understanding of applied economic relationships and for meaningful research in economics. This paper accordingly is devoted to equip the students with basic theory of econometrics and relevant applications of the methods. The topics covered in the course include various problems faced in estimation of both single equations and simultaneous equation models.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one question from each Unit. Each such question would carry 10 marks.

Unit-I

Introduction: Meaning and rationale of Econometric as a separate discipline, Methodology, Types of Econometrics, Nature and Sources of Data used in Econometric Analysis. Functional Forms of Regression Models.

Linear Regression Model: Assumptions, Estimation [OLS], Properties, Interpretation of Estimates and Hypothesis Testing.

UNIT – II

Dummy Variables: Introduction and Uses [as independent variables].

Estimation Problems: Nature, Consequences, Detection and Remedial Measures of the problems of Autocorrelation, Hetroscedasticity, Multicollinearity, and Specification Errors.

UNIT – III

Dynamic Econometric Models: Distributed Lag and Autoregressive Models-Rationale, Problems, Kyock and Polynomial Approaches of Estimation, and Adaptive Expectation and Partial Adjustment Models, Causation and Granger Test.

Simultaneous Equation Models: Nature of Simultaneous Equation Models, Problem and Consequences of Simultaneity in Regression Models. Identification Problem. Derivation of Rank and Order Conditions for Identification of Simultaneous Equations.

UNIT – IV

Estimation of Simultaneous Equation Models: Indirect Least Squares, Two Stage Least Squares, Least Variance Ration, and Three Stage Least Squares Methods of Estimation of Simultaneous Equation Models.

References:

Damodar Gujarati (2003)	:	Basic Econometric [4th Edition], McGraw Hill, New York.
Jack Johnston and John Dinardo (1997)	:	Econometric Methods [4th Edition], McGraw Hill, New York.

OR

Paper – III : Econometrics-II

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hours : 40
Time : 3 Hours

OBJECTIVES: The aim of this paper is to make the students familiar with the use of Limited Dependent Variable models in econometric analysis and also with the application of econometrics to household demand and production analysis in agriculture and industry. problems faced in estimation of both single equations and simultaneous equation models.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with four from each unit and one question spread over the entire syllabus.
The 1st question consisting of 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt two questions from each Unit. Each such question would carry 10 marks.

Unit-I

*Discrete Dependent Variable Mo*dels: Linear Probability Model and its Problems; Logit [both Binomial and Multinomial] and Probit Models: Assumptions; Maximum Likelihood Estimation Methods; Interpretation of Coefficients; Constructing Probabilities; Restrictions and Limitations; Marginal Effects; Measuring Goodness-of-fit; Testing Parameter Restrictions.

UNIT – II

Limited Dependent Variable Models
Truncated and Censored Samples; Sample Selection Bias; The Truncated Regression Model; Marginal Effects; The Tobit Model; Interpretation of Tobit Model Coefficients; Testing for Normality; Limitations of the Tobit Model.

Bivariate Generalisations of the Tobit Model; the Selectivity(Heckit) Model; Two-step and Full-information Estimation Methods; Interpretation of Model Coefficients; Diagnostic Testing; the Double Hurdle (DH) Model; Switching Regressions; Diagnostic Testing.

Unit-III

Econometrics Application to Demand Analysis: Introduction, Single Demand Equations, Systems Demand Equations, Dynamic Demand Analysis.

UNIT – IV

Econometric Applications to Production Analysis in Agriculture and Industry: Basic Introduction to Theory of Firm, Relationship between Production, Cost and Profit Functions, Cobb-Douglas, CES, and Translog Production Functions.

Frontier Production Function and Technical Efficiency. Random Coefficient Frontier Production Function, Cost Function, and Employment Function. Total Factor Productivity- Measurement, Growth, and Decomposition.

References:

Jack Johnston and John Dinardo (1997)	:	Econometric Methods [4th Edition], McGraw Hill, New York.
Michael D. Intriligator (1996)	:	Econometric Models, Techniques, and Applications, (2nd Edition) Prentice-Hall, New Jersey.
K. L. Krishna (1999)	:	Econometric Application in India, Oxford University Press, New Delhi.
William H. Green (2000)	:	Econometric Analysis, (4th Edition) Prentice-Hall, New Jersey

SEMESTER – II

PAPER IV - ECONOMIC THEORY – I

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hours 40
Time : 3 Hours

Objectives: The aim of this paper is to make the students familiar with some of the recent developments in micro and macro economics which have wide range of applicability to solve and explain phenomena; some of which can provide new insights into the empirical estimation of economic relationship.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one question from each Unit. Each such question would carry 10 marks.

UNIT – I

I MARKET FAILURE: SOME MICRO ECONOMIC ASPECTS

A. Externalities: (i) Inefficiency of resource allocation in the presence of externalities; Impact of Market Structure on social welfare loss due to negative externalities.

(ii) Reducing negative externalities. Emission Standards, Emission Fees, Cost & Benefit analysis of reducing negative externalities. Taxing negative externalities in Non Competitive market structures.
(iii) Allocating Property Rights to reduce negative externalities: Coase Theorem; Markets for Pollution.

B. COMMON PROPERTY. Issue of over use and solution.
Public goods. Paretian conditions for Public goods. Revealing preferences and valuing Public goods.

UNIT – II

II. **Markets with Asymmetric Information:** Issues of hidden characteristics/actions and Adverse Selection/Moral Hazard in Insurance and Labor Markets. Response of Market and Government to asymmetric information: efficiency effects.

III. **Questioning the Orthodoxy:** The emergence of Behavioral Economic Reference Points, Adaptation and Losses. Social Preference and Fair allocations; Law of small numbers; Belief Perseverance and confirmatory Bias.

UNIT – III

Monetarist Keynesian debate on Macro theory and Policy.

The Micro-foundations of the Phillips curve

Rational Expectations and the New classical Macro Economics:

- Lucas critique
- Policy ineffectiveness
- Time inconsistency
- Dornbusch's overshooting model
- The random walk model – modern approach to consumption
- Real Business Cycle Theory

UNIT – IV

New Keynesian models of macroeconomic equilibrium:

(a) Real and nominal price rigidity:
Menu costs; coordination failures; imperfect competition.

(b) Real and nominal wage rigidity:
Insider-outsider models, efficiency wages; implicit contracts.

References:

Pindyck, R.S. Rubinfeld, D L & P.L. Mehta (2005)	:	Microeconomics (5th Edition), Pearson Education, Delhi, Chapters 16, 17, 18.5.
Katz, M L and Harvey S Rosen (1999)	:	Microeconomics (3rd Edition) Irwin Chapter 16.
Szenberg, M and Ramrattan L (2004)	:	New Frontiers in Economics, Cambridge University Press. (following readings)

(i) Reading of Behavioral Economics by Mathew Robin

(ii) Information and the change in the Paradigm in Economics by Joseph E Stigliz

Dornbusch, R, S. Fischer, R. Startz (2004) : Macroeconomics, (9th Edition) Tata McGraw Hills Publishing Co. Ltd., New Delhi.

Gupta, G S (2004) : Macroeconomics: Theory and Applications (2nd Edition), Tata McGraw Hills Publishing Co. Ltd., New Delhi.

Eric J. Pentecost (2000) : Macroeconomics: An open economy approach, Macmillan Press Ltd., London.

Bhaduri, A (1993) : Unconventional Economics Essays, Oxford University Press, New Delhi.

Rakshit, M (1989) : Studies in the Macroeconomics of Developing Countries. Oxford University Press, New Delhi.

Patnaik, P (1997) : Macroeconomics, Oxford University Press, New Delhi.

Jha Raghbendra (2008) : Contemporary Macroeconomic Theory and Policy, New Age International Publications.

Paper - V : Economic Theory – II

3 Credits
75 marks
(60+15 internal assessment)
Teaching Hours : 40
Time : 3 Hours

Objectives: The aim of this paper is to make the students familiar with some of the recent developments in micro and macro economics which have wide range of applicability to solve and explain phenomena; some of which can provide new insights into the empirical estimation of economic relationship.

Pedagogy of the Course Work: The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTERS:

Nine questions will be set in all with two from each unit and one question spread over the entire syllabus.
The First question consisting 4 short notes, each carrying 5 marks, set out of the entire syllabi would be compulsory with no internal choice. Additionally, the students are required to attempt one question from each Unit. Each such question would carry 10 marks.

Unit - I

I New growth theory – A Survey
Neo-classical theory of growth and convergence – what it meant; The empirics of growth – what the studies reveal; Explaining the observed patterns of growth through
(a) Increasing returns
(b) Externalities
(c) A K Type models

UNIT - II

II Models of Optimal growth – From Ramsey to McKenzie

(a) The Ramsey Exercise
(b) Philip's golden rule
(c) Discounting the future and Koopman's axiomatization.
(d) Turnpike theorems – an exposition.

Unit – III

Recent Developments in the Theory and Role of Central Banks.
Recent Developments in the Theory of Financial Markets and Asset Pricing.

UNIT – IV

Fiscal Responsibility and Budget Management Act – Issues and Challenges.
Tax and Expenditure Reforms in India.

References:

Romer, David (2005)	:	Advanced Macroeconomics, (3rd Edition) McGraw-Hill, Singapore.
Barro, Robert J & Xavier Sala-I-Martin (2003)	:	Economics Growth (2nd edition), McGraw-Hill, Singapore.
Reddy, Y V (2004)	:	Lectures on Economic and Financial Sector Reforms in India Oxford University Press, New Delhi.
Romer, Paul, M		The Origins of Endogenous Growth, The Journal of Economic Perspectives Volume 8, Number 1, Winter 1994.
Grossman Gene, M & Elhanan Helpman		Endogenous Innovation in the Theory of Growth, The Journal of Economic Perspectives Volume 8, Number 1, Winter 1994.
Solow Robert M	:	Perspective on Growth Theory, The Journal Economic Perspectives Volume 8, Number 1, Winter 1994.
Pack Howard	:	Endogenous Growth Theory: Intellectual Appeal and Empirical Shortcomings, The Journal of Economic Perspectives Volume 8, Number 1, Winter 1994.
Szenberg, M and Ramrattan L (2004)	:	New Frontiers in Economics, Cambridge University Press.

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